2018 JC2 H2 BIO P4 PRELIM EXAM ANSWER SCHEME

Answer **all** questions.

Question 1

A business has been buying milk from the same supplier for a number of months. Recently, the business has found that the milk has been diluted with water. Milk contains macromolecules like proteins which are denser than water thus milk sinks when placed in aqueous solutions.

(a) Predict the behaviour of a milk droplet when placed in water with respect to milk's water content.

milk droplet with more water will sink slower than a drop with less water (ORA); [1]

The amount of water added to a milk sample can be determined by measuring the density of the milk using aqueous solutions like copper sulfate solution of a standard concentration. When a small drop of milk is placed in copper sulfate, a layer of copper proteinate forms around the milk and this prevents the milk and copper sulfate solution mixing.

Fig. 1.1 shows the movement of a drop of milk through the copper sulfate solution.

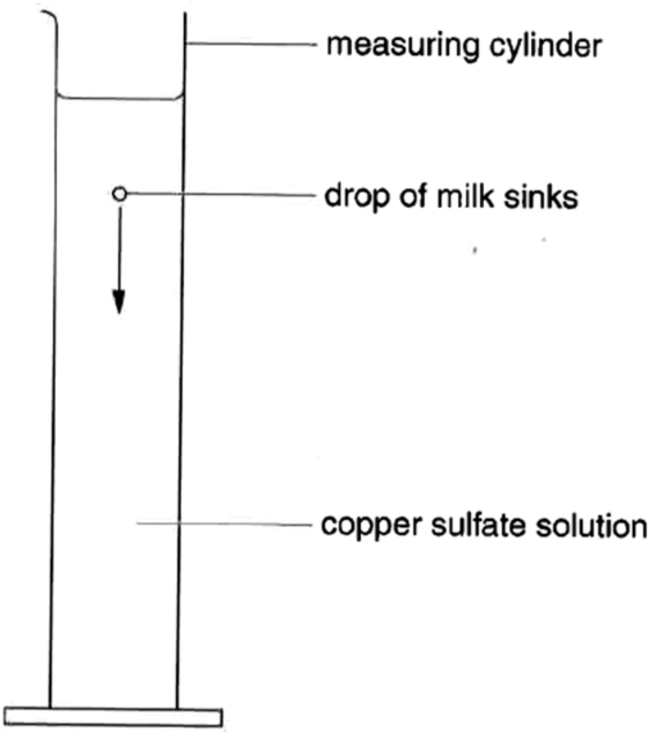


Fig. 1.1

You are required to estimate the percentage of water added to the milk supplied to the business.

You are provided with

- 100% milk, labelled **M**
- milk sample supplied to business, labelled **B**
- distilled water, labelled **W**
- 0.03 moldm^{-3} copper sulfate, labelled **C**

You are advised to read through the entire procedure before beginning the experiment.

1 Prepare 10 cm^3 each of a suitable number of concentrations of milk to help you in your investigation. Record the volume of 100% milk, **M** and distilled water, **W** used in your preparation in a table below.

1. appropriate layout + headings;

2. at least 5 conc.

of wide range, regular interval;

3. correct vol of M & W

total vol of 10 cm^3;

Concentration of M / %	*Volume of M / cm^3*	*Volume of W / cm^3*	*Total Volume / cm^3*
100	*10.0*	*0.0*	*10.0*
80	*8.0*	*2.0*	*10.0*
60	*6.0*	*4.0*	*10.0*
40	*4.0*	*6.0*	*10.0*
20	*2.0*	*8.0*	*10.0*

[3]

2 Using the syringe with attached needle, release one drop of **M** into **C** in a measuring cylinder. Record the time taken by the droplet to sink in an appropriate format in the space provided below.

Note: Needle attached to syringe is sharp. Handle with care. Keep needle capped when not in use.

Observe the largest fragment of **M** should the droplet break up in the copper sulfate solution.

1. layout + headings w units;

2. at least 10 sets of readings

in seconds to nearest whole no.;

3. correct trend;

Suggested table format:

Concentration of M / %	*Time taken by droplet of M to sink (e.g. 100cm³ to 0cm³ mark = 15.6cm) / s*			*Rate / cms^{-1} (Distance/Time)*
	R1	*R2*	*Average [(R1+R2)/2]*	
100				
80				
60				
40				
20				
B				

[3]

3 Repeat **step 2** for all milk concentrations you have prepared in **step 1**. You may reuse the copper sulfate unless the milk residue obstructs your vision.

4 Repeat the procedure to obtain a **total of 2 replicates**. Perform appropriate calculations on your readings. [1]

1. calculates ave time in seconds

to nearest whole no.;

5 Describe how you would carry out **step 2** to increase the accuracy of your observations.

1. use (equal) vol of copper sulfate

of appropriate height;

2. release the milk droplet at the same height / depth

e.g. at the 90 cm^3 mark;

3. release fixed vol of milk e.g. 0.1 cm^3

4. start stopwatch immediately / simultaneously

5. record time for droplet to fall to a fixed point

e.g. reach bottom / e.g. 10 cm^3 mark; [3]

6 Calculate the speed at which the milk droplets sank in distance per unit time. Include your calculations in **step 2**.

[2]

1. calculates speed (correctly) in mm s^{-1} or cm s^{-1};

2. to consistent precision e.g. 3 s.f. or 2 d.p.;

7 Estimate the percentage of water added to the milk sample supplied to business, **B**. Explain how you derived at your answer.

Time taken for a droplet of B to sink = 15 s

Based on table in Step 2, 15s corresponds to 20 – 40% of M, hence % of water added is <u>60 – 80%</u>

percentage of water added ***correct range w units*** [1]

Explanation ***1. release a droplet of B into same vol of copper sulfate at same height***

find time taken for B to sink same distance as rest of milk droplet;

2. find (range) milk conc. that corresponds to time taken by B

100% - % of milk present; [2]

8 Describe **one** way to improve your estimate in terms of

(a) reliability;

1. 3 replicates / 2 repeats

to calculate ave to eliminate random error OR identify anomaly; [1]

® ensure reproducibility

(b) accuracy.

1. repeat procedure using milk conc. of smaller intervals

within range identified in step 6; [1]

® using smaller intervals of irrelevant milk conc.

Further investigation was conducted to find the protein concentration in sample **B** using biuret's test. The absorbance by sample **B** was measured using a colorimeter and compared to a range of protein solutions of known concentrations. Table 1 shows the absorbance by the protein solutions.

Table 1

Protein concentration / %	Absorbance / arbitrary units
100	65
80	55
60	38
40	21
20	10

9 Describe how you would conduct biuret's test using 1 cm^3 of milk sample.

1. add equal vol / 1 cm^3 of biuret soln to milk sample;

(mix with glass rod)

2. violet / purple / lilac colouration indicates presence of prot; [2]

10 Plot a suitable graph using data provided in Table 1.

1. correct axes (indep var on y-axis, dep var on x-axis)
axes labels with units;

2. appropriate scale (at least half of grid, able to estimate to half of smallest square)
correct data pts;

3. best fit (straight line with data pts evenly distributed on both sides / point-to-point plot)
does not extrapolate graph;

[3]

11 The absorbance of the milk sample **B** was recorded to be 26 arbitrary units. Using your graph, deduce the protein concentration in the milk sample **B**. Show on your graph, how your arrived at your answer.

value in % e.g. 42.5%;

protein concentration of milk sample, **B** ***show on graph ;*** [2]

[Total: 25]

Question 2

You are provided with a microscope slide of a cross section of monocotyledonous ster **K1**. Fig. 2.1 shows an outline of the stem.

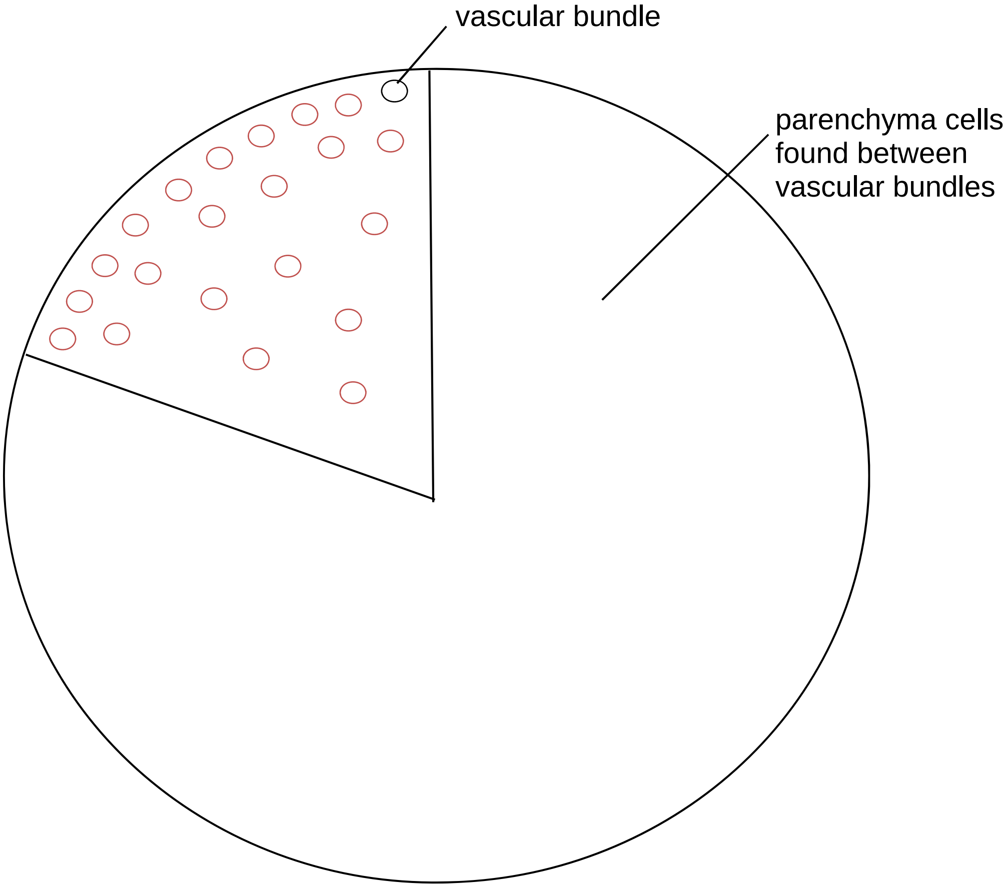


Fig. 2.1

1 View **K1** using the **X4 objective** lens. Indicate the distribution of vascular bundles in one fifth of the stem in Fig. 2.1 using open circles, O. One vascular bundle is shown in Fig. 2.1. [2]

1. correct relative size ;

2. more near epidermis ;

2 You are provided with a stage micrometer. Each division measures 0.01mm. Using the stage micrometer, estimate the diameter of the field of vision under **X40 objective** in **mm**.

diameter of field of vision ***0.46 mm*** [1]

3 Using your answer in step 2, calculate the diameter of the field of vision under **X10 objective** in **mm**. Show your working.

Diameter of field of vision is inversely proportional to the viewing magnification, therefore, diameter in X10 = 0.46 mm x4
= 1.84 mm

diameter of field of vision ***1.84 mm*** [1]

4 Count the number of vascular bundle in the field of vision viewed using **X10 objective** lens. Avoid the centre of the specimen.

number of vascular bundles ***e.g. 9 - 18*** [1]

5 Calculate the density of vascular bundles in **K1** using the formula

$$\text{Density of vascular bundle} = \frac{\text{Number of vascular bundle}}{\text{Area of field of vision}}$$

Show any other calculations you performed to obtain the answer.

Radius of FOV = 1.84mm/2
= 0.92 mm

Area of FOV = πr^2
= $2.66\ mm^2$

Density = 10 / 2.66
= $3.76\ mm^{-2}$

density of vascular bundles ***in*** mm^{-2} [1]

6 Comment on the reliability of your answer in **step 5**.

1. not reliable

as only 1 FOV observed / distribution of vascular bundles are varied;

2. take at least 3 readings to obtain average OR

count whole stem / count diff parts with diff distribution; [2]

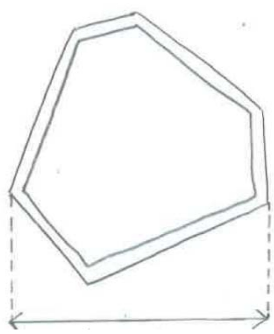
7 View a typical vascular bundle using high power objective. State the objective lens you used.

objective lens ***X40 or X60*** [1]

8 Draw, **to the same scale**, **one** typical cell from each of the following type

(a) phloem

1. angular shape;
2. thin cell wall;
3. continuous lines;



15μm (at x40 & x60)

Actual size under x40 objective
= 0.0025mm x 6 divisions
= 0.015mm

****1mm = 1000µm***
0.015mm = 15µm

Actual size under x60 objective
= 0.00167mm x 9 divisions
= 0.015mm

****1mm = 1000µm***
0.015mm = 15µm

Magnification = $\frac{\textit{drawing size}}{\textit{actual size}}$

$= \frac{46\ mm}{0.015\ mm}$

$=$ X 3066.67

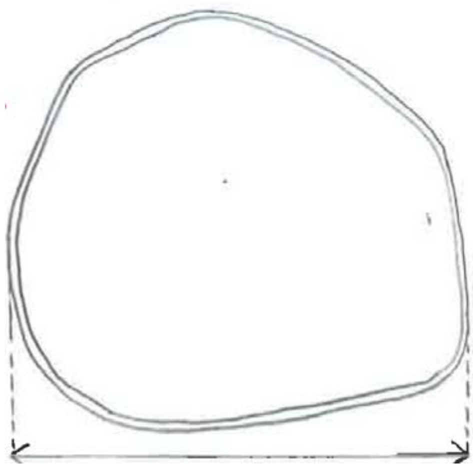
Answer = ***~ X 3066.67*** [3]

(b) parenchyma

1. round shape;

2. thinner cell wall than phloem;

3. continuous lines;



~ 62.5µm (at x40 & x60)

Actual size under x40 objective
= 0.0025mm x 25 divisions
= 0.0625mm

****1mm = 1000µm***
0.0625mm = 62.5µm

Actual size under x60 objective
= 0.00167mm x 38 divisions
= 0.0625mm

****1mm = 1000µm***
0.0625mm = 62.5µm

Magnification = $\frac{\textit{drawing size}}{\textit{actual size}}$

$= \frac{92\ mm}{0.0625\ mm}$

$=$ X 14720

Answer = ***~ X 1472*** [3]

9 **(a)** Indicate the **actual size** of the cells drawn in **8(a)** and **(b)**, in **µm**. [2]

(b) Calculate the magnification of your drawings and show your working clearly in **8(a)** and **(b)**.

calculates magnification = drawing size / actual size;

[2]

[Total: 19]

Question 3

Many people are intolerant to the disaccharide lactose, which is found in milk. The enzyme lactase is used commercially to catalyse the breakdown of lactose to the monosaccharides glucose and galactose. These sugars taste sweeter and are easier to digest than lactose.

Enzymes can be immobilised in a number of different ways, using different materials.

Fig. 3.1 shows three ways of immobilisation of enzymes.

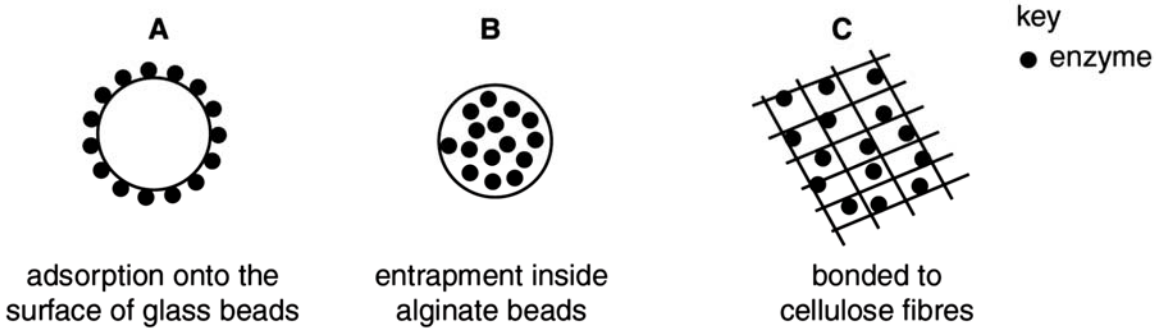


Fig. 3.1

A student carried out an investigation to compare the activity of the enzyme lactase that had been immobilised in the three different ways shown in Fig. 3.1. The enzyme can be reacted with lactose by pouring lactose solution through a column containing the immobilised enzyme. The subsequent glucose produced can be measured using a biosensor (Fig. 3.2) that can measure the concentration of glucose in a solution in mg/100cm^3.

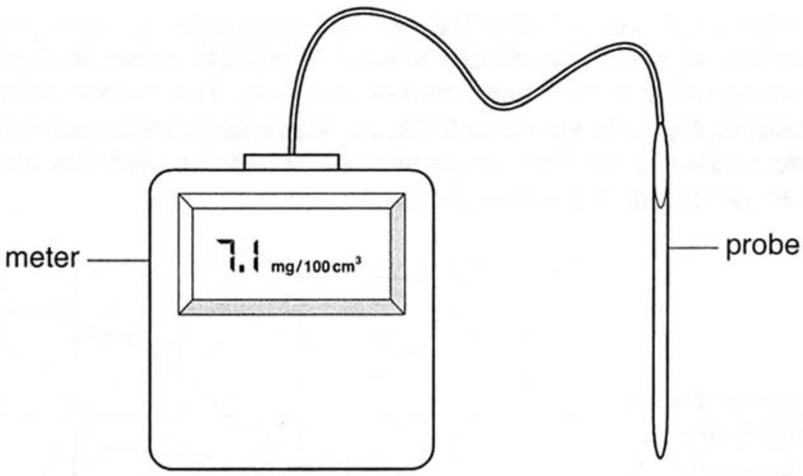


Fig. 3.2

Outline a procedure the student could use to compare the activity of lactase that has immobilised in different ways.

You are provided with and must use:

- lactose solution
- lactase immobilised using glass beads, **A**
- lactase immobilised using alginate beads, **B**
- lactase immobilised using cellulose fibres, **C**
- biosensor that measures glucose concentration
- stopwatch

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- apply appropriate statistical test to support conclusion,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 11]

IV	*1.*	*method of immobilisation;*
DV	*2.*	*amount of product produced*
		measured glucose conc. using biosensor;
CV	*3.*	*temp, specify < 60^{o}C,*
		temperature controlled room / incubator / water bath;
CV	*4.*	*pH*
		pH 7, using pH buffer;
P	*5.*	*constant conc. and vol of lactase*
		constant conc. and vol of lactose;
	6.	*fixed time allowed for rxn, using adjustable clip ® constant pour rate*
		timed with stopwatch;
	7.	*means of collecting product;*
	8.	*apparatus set up (A) in diag;*
R+R	*9.*	*3 replicates, 2 repeats*
		calculate ave / identify anomaly;
R	*10.*	*results table, layout, headings, units;*
	11.	*calculates rate in mg/100cm^{-3} per unit time;*
	12.	*state Ho;*
	13.	*use t-test on diff pairs of ave*
		$p < 0.05$ as significant;
G	*14.*	*bar graph;*
S	*15.*	*state hazard, risk, precaution*
		e.g. lactase, skin irritant, wear gloves / goggles;